



江苏省苏州中学国际部
Suzhou High School – International Division

SZZX-ID Bridge Program Physics

Unit 3 Lesson 4: Newton' s law of gravitation – virtual lab

Class Outline

Title of Unit	Projectile and circular motions
Title of Lesson	Newton's law of gravitation
Starter Activity	Introduce the universal gravitation
Direct Instructions	Teaching Strategies: Demonstration, Lecture, Collaboration, Questioning, Inquiry
	Lecture: The development of the universal law of gravitation. The mathematical expression. Inquiry: calculate the mass of the Sun; gravitational virtual lab
	Differentiation: Content, Instructional Strategies, the Classroom, Products, Teacher
	Instructional strategies: give support during the virtual lab. Template for data provided
	International Mindedness:
Lesson Objectives	By the end of the lesson, students will be able to: • Understand the expression of the universal gravitational force between two objects • Understand how the law of universal gravitational force was used to describe the motion of satellites, planets, moons etc • Skill: Record data and plot graphs digitally
ATLs	Identify unit ATLs and how they are being addressed in the lesson
	Communication Social
IB Learner Profile	Link the lesson to the characteristics of the IB Learner attribute
	Inquirers: We nurture our curiosity, developing skills for inquiry and research.
Teaching Activities	• Activity: calculate the mass of the sun • Practical: gravitation virtual lab
Plenary Session	Checking for understanding / Homework / Q & A / Reflection etc.
	Practice using calculator

Vocabulary



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- Proportional 成比例的
- Product 乘积
- Universal 通用的
- Universal gravitation 万有引力
- Orbit 轨道/绕轨道运行
- Virtual 虚拟的
- Investigate 调查、研究

Newton's law of gravitation



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- In 1687 Isaac Newton published what has been called by some the greatest scientific discovery of all time – his universal law of gravitation.
- The law states that the gravitational force between two masses m_1 and m_2 is proportional to their product, and inversely proportional to the square of their separation r .

$$F = \frac{Gm_1m_2}{r^2}$$

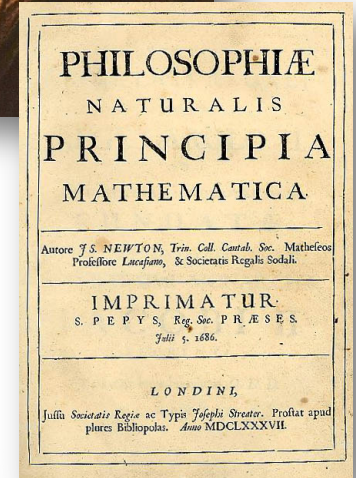
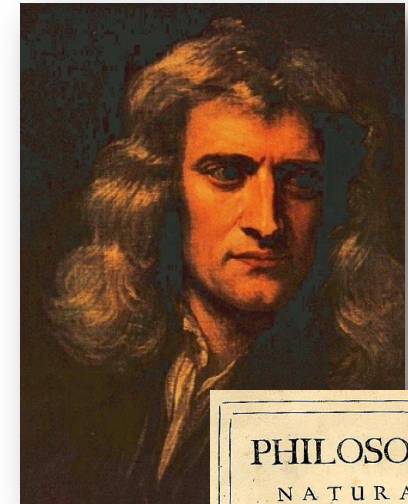
where

$$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$$

G : Universal gravitational constant

Universal law
of gravitation

- All objects with mass attract other objects with mass



Let's calculate the mass of the Sun!



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- The average distance between the Sun and the Earth is 149.6×10^6 km
- The gravitational force between the Sun and the Earth provides centripetal force for Earth's orbiting
- Can you estimate the mass of the Sun?

$$F = \frac{Gm_1m_2}{r^2}$$

where

$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
 G : Universal gravitational constant

Universal law
of gravitation



Virtual lab



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- Aim

To investigate how the mass of one planet and the distance between two planets affect the gravitational force between the two planets.

- Tool

<http://phet.colorado.edu/en/simulation/gravity-force-lab>

- Skills

Record data and plot graphs digitally

